DocuMind Academic Benchmark: Quantum Search Algorithms

Abstract:
Quantum computation exploits superposition and entanglement to solve unstructured search problems. We revie

1. Foundational Principles:
A qubit resides in a state |psi> = alpha|0> + beta|1>, where |alpha|^2 + |beta|^2 = 1. Superposition allows 2^n sir

2. Grover's Algorithm Analysis:
Grover's algorithm provides quadratic speedup for searching an unsorted database of N items. While classical se

3. Physical Implementations and Fault Tolerance:
Physical realization of Grover's search was demonstrated using superconducting transmon qubits. Experimental